Explanation of the Paper and Implementation

1 Problem

The paper focuses on the problem of improving text classification models when the available training data is limited or not diverse enough. In many natural language processing tasks, especially for low-resource languages, collecting large labeled datasets is difficult. Because of this, models may not learn enough useful patterns and may perform poorly on unseen data. A common solution to this problem is data augmentation, where new training examples are created from existing examples.

2 Solution

The paper proposes an Explainable AI-guided data augmentation method. The method first uses an XAI technique to identify which words are less important for the model's prediction. These less important words are then modified using two augmentation methods: synonym replacement with back-translation and paraphrasing with back-translation. The goal is to preserve the original meaning and label of the sentence while creating new sentence variations that can improve the model's performance.

3 Implementation

In my implementation, I followed a simplified version of the pipeline. I used two transformer-based multilingual models, XLM-R and mBERT, for both hate-speech detection and sentiment analysis as baseline models. For each dataset, a subset of 1000 training samples and 250 validation samples was used. These smaller subsets were chosen because using larger datasets caused runtime crashes due to computational limitations. Since the models, Integrated Gradients calculations, translation, paraphrasing, and retraining steps are computationally expensive, using the full datasets was not possible.

First, the baseline models were trained and evaluated. Then, Integrated Gradients was used to estimate the importance of the words in each sentence. The least important words were selected for modification. After that, augmented datasets were created using XAI-SR-BT and

XAI-PR-BT. Finally, the augmented data was combined with the original training data, and the models were retrained. The accuracy and F1 scores after augmentation were then compared with the baseline results.

4 Results

For the hate-speech results, the augmentation methods did not generally improve performance. For XLM-R, the results stayed exactly the same for Italian, Hindi-English, Portuguese, and Spanish. This means that the added augmented samples did not change the model's final predictions on the validation set. For mBERT, the Hindi-English and Portuguese results also stayed the same, but Italian performance decreased after both augmentation methods, and Spanish performance decreased after XAI-SR-BT. This suggests that, for hate-speech detection, the augmentation methods may not have produced useful new examples, or they may have introduced noisy examples that made training worse.

For the sentiment-analysis results, most results also stayed the same or decreased. XLM-R did not improve on any language. In Hausa, XAI-PR-BT slightly reduced the score, and in Kinyarwanda both augmentation methods caused a large decrease. For mBERT, Amharic, Hausa, and Swahili stayed unchanged. The best result appeared in Kinyarwanda, where mBERT improved from 0.200 accuracy and 0.156 F1 before augmentation to 0.600 accuracy and 0.506 F1 after XAI-SR-BT. XAI-PR-BT also improved Kinyarwanda compared with the baseline, but not as much as XAI-SR-BT.

Overall, the results show that the XAI-guided augmentation method did not consistently improve performance across all datasets and models. The strongest positive result was seen for mBERT on Kinyarwanda sentiment analysis, where augmentation clearly helped. However, many results stayed unchanged, and some decreased after augmentation. One important reason for this difference from the original paper is that the experiments used smaller subsets because of computational limitations. With smaller subsets, the models may not learn as robustly, and the validation scores can be more unstable. In addition, the translated or paraphrased words may not always have preserved the intended meaning, and some augmented examples may have introduced noise.

Table 1: Hate speech results

Model	Language	Before augmentation		XAI synonym with back translation				XAI paraphrasing with back translation			
		Acc	F1	Acc	F1	ΔAcc	ΔF1	Acc	F1	ΔAcc	ΔF1
XLM-R	Italian	0.542	0.381	0.542	0.381	0.000	0.000	0.542	0.381	0.000	0.000
	Hindi-English	0.867	0.805	0.867	0.805	0.000	0.000	0.867	0.805	0.000	0.000
	Portuguese	0.900	0.853	0.900	0.853	0.000	0.000	0.900	0.853	0.000	0.000
	Spanish	0.585	0.432	0.585	0.432	0.000	0.000	0.585	0.432	0.000	0.000
mBERT	Italian	0.617	0.611	0.542	0.381	-0.075	-0.231	0.542	0.381	-0.075	-0.231
	Hindi-English	0.867	0.805	0.867	0.805	0.000	0.000	0.867	0.805	0.000	0.000
	Portuguese	0.900	0.853	0.900	0.853	0.000	0.000	0.900	0.853	0.000	0.000
	Spanish	0.585	0.432	0.415	0.243	-0.170	-0.188	0.585	0.432	0.000	0.000

Table 2: Sentiment analysis results

Model	Language	Before augmentation		XAI synonym with back translation				XAI paraphrasing with back translation			
		Acc	F1	Acc	F1	ΔAcc	ΔF1	Acc	F1	ΔAcc	ΔF1
XLM-R	Amharic	0.500	0.333	0.500	0.333	0.000	0.000	0.500	0.333	0.000	0.000
	Hausa	0.350	0.181	0.350	0.181	0.000	0.000	0.300	0.138	-0.050	-0.043
	Kinyarwanda	0.600	0.512	0.400	0.229	-0.200	-0.283	0.400	0.229	-0.200	-0.283
	Swahili	0.600	0.450	0.600	0.450	0.000	0.000	0.600	0.450	0.000	0.000
mBERT	Amharic	0.500	0.333	0.500	0.333	0.000	0.000	0.500	0.333	0.000	0.000
	Hausa	0.350	0.181	0.350	0.181	0.000	0.000	0.350	0.181	0.000	0.000
	Kinyarwanda	0.200	0.156	0.600	0.506	0.400	0.351	0.400	0.229	0.200	0.073
	Swahili	0.600	0.450	0.600	0.450	0.000	0.000	0.600	0.450	0.000	0.000